^{36}Ar(p,t) 2018Lo11,1972Pa02,1969Br21

^{36}Ar→2n+^{34}Ar from J^{π}=0^{+} ^{36}Ar ground state.

2018Lo11: a 100-MeV proton beam was produced from the RCNP WS beam line at Osaka University. The target was 0.76-mg/cm^2 enriched ^{36}Ar gas (≥99%). Reaction products were momentum-analyzed by the Grand Raiden spectrograph and detected using the focal-plane detector system. Measured σ(E$_t$,θ). Deduced levels. Calculated astrophysical ^{30}S(α,p)^{33}Cl reaction rate. Also see 2009ObZY.

1972Pa02: 40-45-MeV proton beams were produced from the MSU sector-focused cyclotron. The target was >99%−enriched ^{36}Ar gas. Reaction products were detected using a detector telescope made up of two silicon surface-barrier transmission mounted counters. Measured σ(E$_t$,θ). Deduced levels, L-transfers, J, and π from JULIE-DWBA analysis.

1969Br21: a 45-MeV proton beam was produced from the Berkeley 88-Inch Cyclotron. The target was 99.6%-enriched ^{36}Ar gas. Reaction products were detected using a 155−υm phosphorus-diffused silicon ΔE transmission counter and a 3.0-mm lithium-drifted silicon E counter operated in coincidence, followed by a 0.5-mm lithium-drifted silicon E-reject counter operated in anti-coincidence with the first two counters to eliminate long-range particles. Measured σ(E$_t$,θ). Deduced levels, L-transfers, J, and π from DWUCK-DWBA analysis. Also measured ^{36}Ar(p,^{3}He)^{34}Cl and ^{36}Ar(d,α)^{34}Cl.

1974Ha02: a 40.2-MeV proton beam was produced from the MSU cyclotron. Measured Q value. Tritons for the g.s. to g.s. transition measured using split-pole magnetic spectrograph, FWHM=10-25 keV. Measures Q value=−19513 *3*. Deduced mass excess of ^{34}Ar.

^{34}Ar Levels

E(level)†	J^{π}	L‡	Comments
0	0$^+$	0	
2091.6 *24*	2$^+$	2	E(level): weighted average of 2094 *11* (1972Pa02) and 2091.5 *24* (2018Lo11). Other: 2097 *20* (1969Br21). L: 2 in 1972Pa02.
3290.9 *18*	2$^+$	2	E(level): weighted average of 3303 *25* (1969Br21), 3288 *14* (1972Pa02), and 3290.9 *18* (2018Lo11). L: 2 in 1972Pa02.
3879.1 *33*	0$^+$	0	E(level): weighted average of 3899 *25* (1969Br21), 3879 *15* (1972Pa02), and 3878.8 *33* (2018Lo11). L: 0 for unresolved 3879+4050 in 1972Pa02; 3879 is more strongly excited.
4035 *16*			E(level): unweighted average of 4050 *14* (1972Pa02) and 4019.1 *43* (2018Lo11).
4515.1 *23*	(3$^-$)	(3)	E(level): weighted average of 4522 *14* (1972Pa02) and 4514.9 *23* (2018Lo11). Other: 4560 *40* (1969Br21). L: not fitted particularly well by either L=2 or L=3 calculated distributions. However, it is distinctly different from the other experimental L=2 distributions. There is also a 3$^-$ state at approximately the same excitation energy in the mirror nucleus ^{34}S (1969Br21).
4641.5 *21*			E(level): weighted average of 4651 *14* (1972Pa02) and 4641.3 *21* (2018Lo11).
4875.3 *38*			E(level): weighted average of 4867 *14* (1972Pa02) and 4875.9 *38* (2018Lo11).
4967.9 *27*	0$^+$	0	E(level): weighted average of 4970 *40* (1969Br21), 4985 *14* (1972Pa02), and 4967.2 *27* (2018Lo11).
5262 *15*			
5317 *13*			E(level): weighted average of 5340 *40* (1969Br21), 5307 *13* (1972Pa02), and 5330 *17* (2018Lo11).
5535 *18*			
5629.6 *45*			
5929 *20*			E(level): unweighted average of 5909 *12* (1972Pa02) and 5948.2 *66* (2018Lo11).
6050.1 *36*	(2$^+$)	(2)	E(level): weighted average of 6074 *11* (1972Pa02), and 6049.2 *20* (2018Lo11). Other: 6100 *40* (1969Br21). L: the total angular distribution of unresolved 5909+6074 does exhibit some L=2 character (1972Pa02). L=2 is assigned for 6100 in 1969Br21.
6525 *9*	2$^+$	2	
6641 *14*			
6723 *22*			
6807 *4*			E(level): weighted average of 6794 *11* (1972Pa02) and 6808 *3* (2018Lo11). Other: 6860 *40* (1969Br21).
6917 *5*			
7072 *8*			
7276 *2*			
7340 *11*	2$^+$	2	E(level): unweighted average of 7340 *50* (1969Br21), 7322 *6* (1972Pa02), and 7358 *4* (2018Lo11).
7502 *16*	(2$^+$)	(2)	E(level): unweighted average of 7530 *50* (1969Br21), 7499 *4* (1972Pa02), and 7476 *2* (2018Lo11).
7921 *18*			E(level): unweighted average of 7950 *50* (1969Br21), 7925 *5* (1972Pa02), and 7889 *2* (2018Lo11).

Continued on next page (footnotes at end of table)

^{36}Ar(p,t) 2018Lo11,1972Pa02,1969Br21 (continued)

^{34}Ar Levels (continued)

E(level)†	E(level)†	E(level)†	E(level)†
8166 *3*	8746 *8*	9004 *4*	9226 *18*
8660 *9*	8899 *8*	9148 *22*	9549 *16*

† From 2018Lo11, unless otherwise noted.
‡ From 1969Br21, unless otherwise stated.